# JUAN ESTEBAN ÁLVAREZ GARCÍA

Systems and Computing Engineering | Software development · Cloud/DevOps · Applied security

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Full-time · On-site or hybrid in Bogotá · Remote worldwide

## PROFESSIONAL PROFILE

Trilingual Systems and Computing Engineer from Universidad de los Andes. Hands-on experience in software design, full-stack development and deployment, automation, and cloud services on GCP and AWS. Particular interest in cryptography, defensive cybersecurity, backend/frontend, DevOps, and artificial intelligence. I have built solutions in agile and multicultural teams, applying software architecture, testing, Design Thinking, and a human-centered approach. My strengths include analytical thinking, initiative, effective communication, and the ability to collaborate and lead; with an academic exchange in Germany and certified C1 English.

## EXPERIENCE

**Freelance Developer** | Jul. 2024 - present

- Automated data extraction and the entry of hundreds to thousands of hotel reservations per run with Python/Selenium, reducing a weekly process to approximately one day.
- Designed and launched a dental website (Next.js, Tailwind CSS, Netlify) with a WhatsApp CTA and contact form; it has helped attract ~2-3 new clients per month.

**Teaching Assistant | Universidad de los Andes** | Jan. 2026 – Dec. 2026

- Supported ~60 students in Product Design and Innovation in IT by assessing submissions, answering questions, running support workshops, and creating teaching materials.

**Private Programming Tutor** | Jan. 2022 – Dec. 2026

- I have tutored ~20 students in programming, OOP, data structures, and algorithms; designed exercises and led personalized one-on-one and group sessions of up to four people.

## EDUCATION

**Universidad de los Andes | Systems and Computing Engineering** Jan. 2022 – Dec. 2026 · Expected graduation: 2027 · GPA: 4.2/5.0

**Relevant coursework:** Cryptographic Engineering and Its Applications in IT; Quantum Computing Algorithms; Computational Infrastructure; Design and Analysis of Algorithms; OOP; Data Structures and Algorithms; Fundamentals of Machine Learning; Deep Learning Analysis; Team Software Development; Mobile Application Development; Cloud Software Development; DevOps and Continuous Deployment; Software Architecture and Design.

**Hochschule München | Academic exchange** Munich, Germany · Feb. - Jul. 2025 · 17 ECTS. Software Engineering, Blockchain, Human-Computer Interaction and UX with eye tracking; work in international teams.

**Additional training:** Google GEAR Academy (in progress).

## TECHNICAL SKILLS

**Languages:** Python, Java, TypeScript, SQL, NoSQL, Dart.
**Frontend and mobile:** Next.js (React), Angular, Flutter, Tailwind CSS, Swing.
**Backend and APIs:** FastAPI, Spring, Django, Flask; REST API design and integration.
**Cloud (GCP / AWS):** Cloud Run, Compute Engine, Cloud SQL, Cloud Storage, IAM, Artifact Registry, BigQuery; EC2, ECS/Fargate, Elastic Beanstalk, S3, RDS/Aurora, ELB, Secrets Manager, CloudWatch, CodeBuild and CodePipeline.
**DevOps and tools:** Git, GitHub Actions, CI/CD, Docker, Docker Compose, Linux, Nginx, Jenkins, SonarQube, SSH.
**Data, AI and security:** PostgreSQL, MySQL, MongoDB; Keras, Machine Learning; AES, RSA, TLS, JWT, ZKP, LWE, Kyber and ECDSA.
**Engineering and methodologies:** Software architecture, data structures, Big O analysis, algorithm optimization, automated testing, Agile/Scrum, Design Thinking, ER modeling, TCP/IP, UDP, DNS and networking.

## INTERPERSONAL SKILLS

Analytical problem-solving · Clear and multicultural communication · Collaboration and leadership · Autonomy and initiative · Attention to detail · Working under pressure · Adaptability and user focus · Mentoring and teaching .

## FEATURED PROJECTS

### Thesis | Post-quantum zero-knowledge authentication

*Python · SageMath · LaTeX · Research in progress* | Repository

- Co-designed a quantum-resistant Zero-Knowledge authentication protocol based on Learning with Errors (LWE); I lead the LWE mathematical component.
- Theoretical framework completed; development of benchmarks for time, memory, network traffic, and security estimation in bits.

### AndeSpace | University space reservations

*Flutter · FastAPI · PostgreSQL · GitHub Actions* | Flutter · Backend · Wiki

- Implemented the search, home, reservations, and profile/friends modules end to end, integrating the mobile interface, API, and data.
- Best project in its Mobile Application Development section; code quality supported by SonarQube.

### HabiCapital | Secure digital banking simulation

*Python · FastAPI · Next.js · PostgreSQL · Docker* | Code

- Individually designed and built a P2P platform with applied security: JWT, Argon2id, AES-256-GCM, HMAC, an immutable ledger, and idempotency.
- AI-assisted development with specifications (SDD) and TDD; more than 100 successful automated tests and ~94% reported coverage.

### Teaching assistantship tracking platform | Cloud

*GCP · GitHub Actions · CI/CD*

- Developed a weekly tracking system for instructors and teaching assistants as part of a team; worked with Cloud Run, Compute Engine, Cloud SQL, Cloud Storage, IAM, and Artifact Registry.

### Email blacklist microservice | DevOps

*AWS · GitHub Actions · CI/CD*

- Contributed to developing and delivering a microservice with automated integration and deployment, using ECS/Fargate, RDS, CloudWatch, CodeBuild, and CodePipeline.

### AhorrApp | Personal finance for university students

*Angular · Digital product · UX research* | Frontend · Backend · Landing

- Developed the functional application and participated in interviews, JTBD/ICP definition, iterations with users, and product validation; best project in its section.

### DogSPA | Branch and service management

*Angular · Spring · MySQL*

- Implemented full-stack modules for branches/cities and the service catalog, integrating frontend and backend; best final project in Team Software Development.

## RECOGNITION AND LEADERSHIP

Recognitions in Product Design, Mobile Applications, and Team Software Development; two second-place finishes in Design and Analysis of Algorithms; Data Structures Hall of Fame (2022). Former co-representative of the calisthenics team (tournaments and activities, 2022-2024).

## LANGUAGES

Spanish: native · English: C1 (IELTS General Training 7.5, 2025) · German: B1 (everyday conversation).